IT-SIMPLERA INSTITUTE

INTERNSHIP WEEKLY REPORT

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- Domain: AI/ML
- Role: AI/ML Intern
- Week: 01
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Project Introduction:

- Introduction to EDA:

This topic is about Exploratory Data Analysis (EDA).

In simple words, it means exploring our data using charts and graphs to see what stories the numbers are telling us.

- Introduction to Data Preprocessing:

This topic is also about Data Preprocessing.

Preprocessing simply means cleaning the data.

Real-world data is always messy, has missing values, and contains mistakes. We must clean it up before we can build any charts or machine learning models.

Task Overview:

Our Dataset:

We are working with a huge online retail dataset that contains 525,461 rows of customer shopping history.

Our Goals for this Week:

Cleaning: Finding and removing missing values in the dataset.

Fixing Formats: Changing raw date text into a clean time format so we can track monthly trends.

Feature Engineering: Multiplying item quantities by their prices to calculate the total money (Revenue) made from each sale.

Our Tech Stack:

- We write all of our cleaning and graphing code using Python inside a Jupyter Notebook.
- We use libraries like **Pandas** and **Numpy** to clean the data, and **Matplotlib** and **Seaborn** to draw the graphs.

Executive Summary:

Project Purpose: This report covers our first weekly milestone for the data science internship.

The main goal is to take a large, raw dataset and prepare it so we can find useful business patterns.

We successfully processed a massive retail dataset containing over 525,000 sales records.

We cleaned out the missing information, fixed the date formats, and calculated the total revenue for every transaction.

We created clear charts (line plots, heatmaps, and box plots) to safely find data mistakes and extreme wholesale values.

Final Result:

The data is now clean, correct, and perfectly ready for building machine learning models next week.

Data Preprocessing & Initial EDA:

Data Loading & Inspection:

We loaded our raw dataset into Python to check its size.

The dataset is huge, containing exactly **525,461** rows and **8** columns of data.

Cleaning the Data:

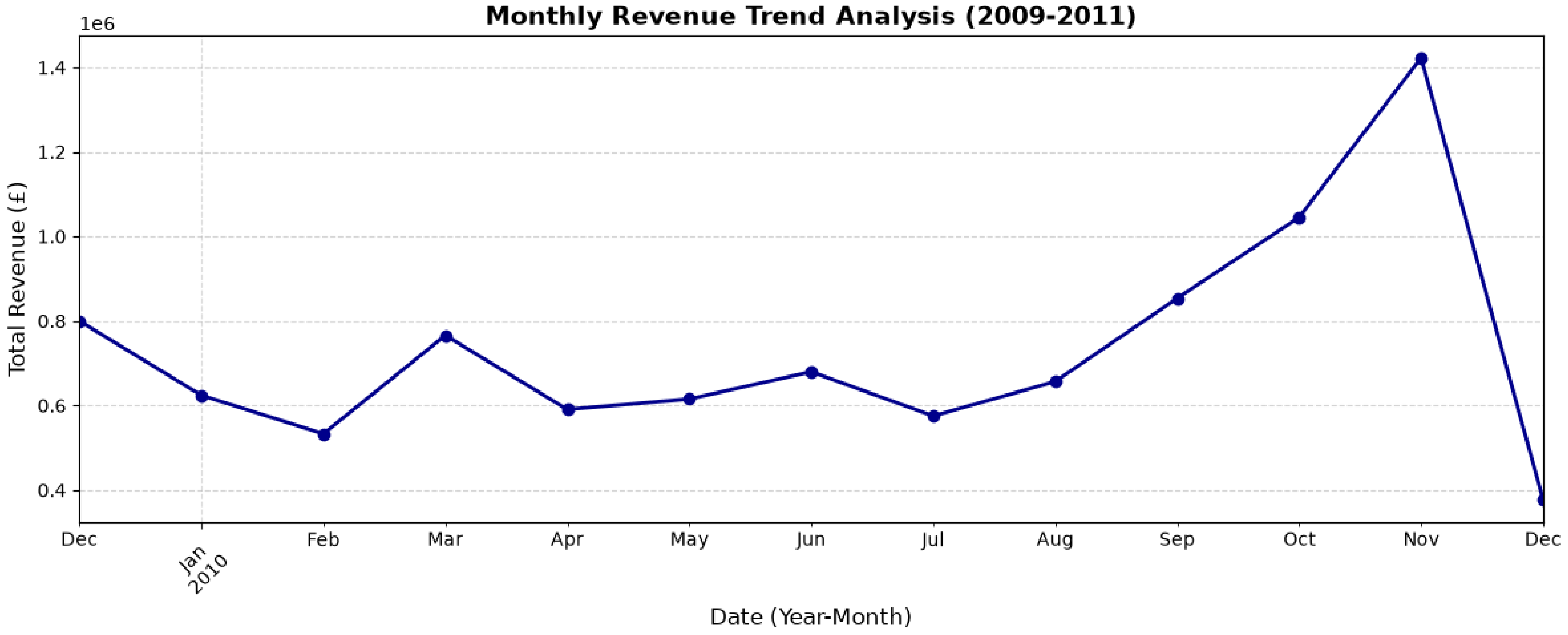
Removing Blanks: We found missing values in the data (like missing Customer IDs) and removed them so our data stays accurate.

Fixing Dates: We changed the raw date text into a clean date format so Python can easily understand timelines.

We also multiplied the Quantity of items by their Price to create a brand-new column called Revenue.

This tells us exactly how much money was made on every single sale.

Monthly Revenue Trend Analysis:



The above slide shows a Line Chart that tracks how much total money (Revenue) the retail store made during each month of the year.

It helps us look at sales trends over time to see which months are the busiest.

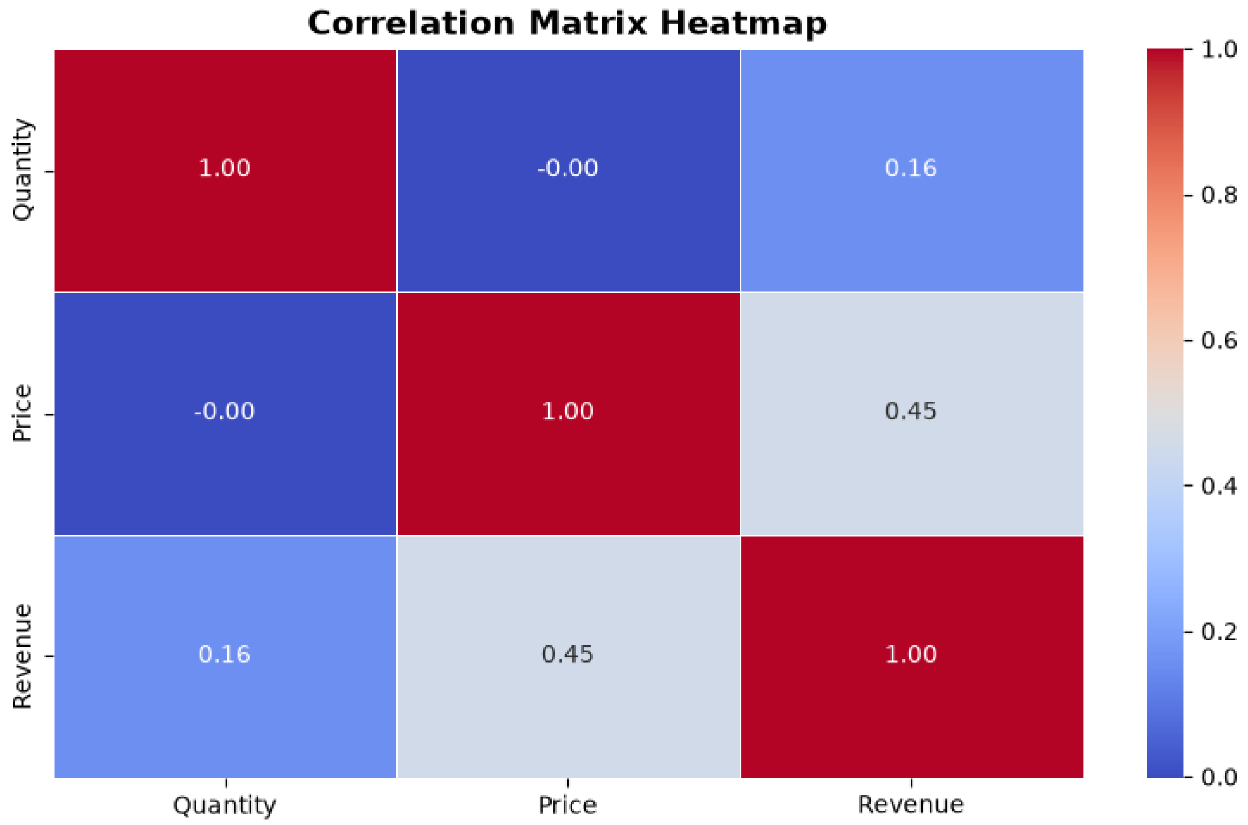
- ## Key Findings from the Chart:

Sales Growth: The graph shows a steady and massive increase in total revenue as the months go by.

Peak Season: Sales velocity peaks heavily during the late winter weeks, showing the store's highest profit season.

Year-End Drop: Right at the very end of the calendar cycle, the chart shows a sharp drop-off in transactions.

Attribute Correlation Heatmap:



The above slide shows a Heatmap chart.

It measures how different numerical columns (Quantity, Price, and Revenue) relate to each other.

The numbers on the chart tell us if two things increase together or if they have no connection.

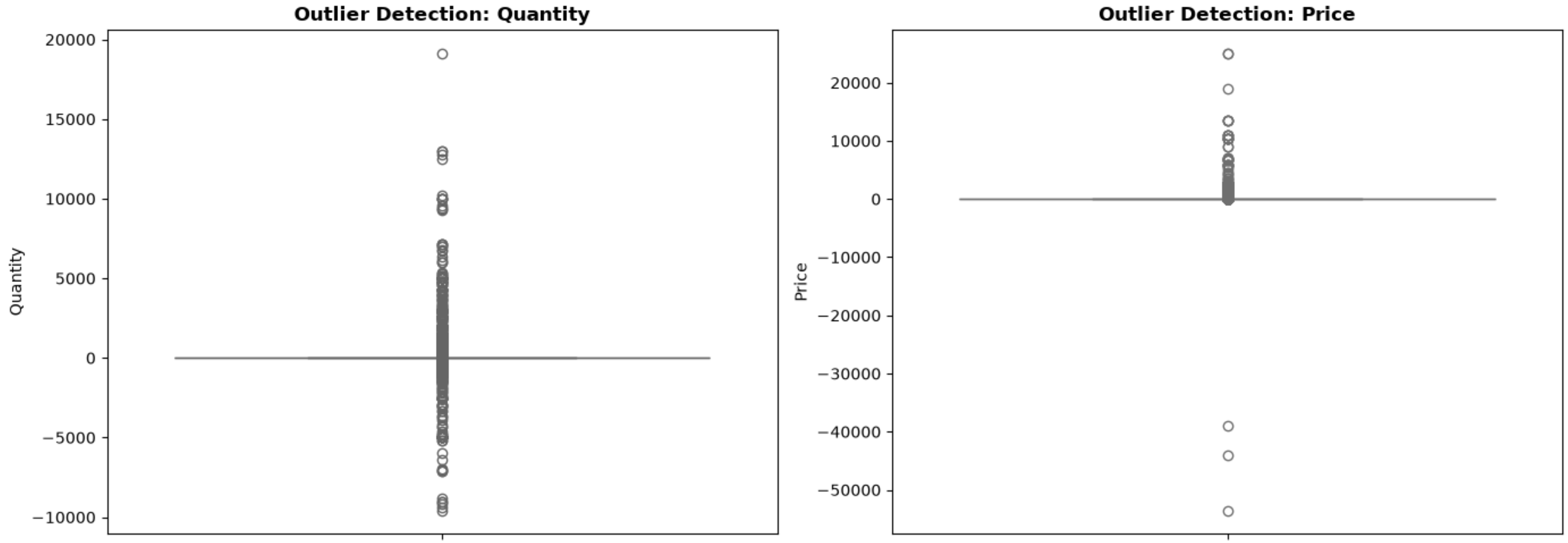
- Key Findings from the Chart:

Feature Interactions: The graph shows that individual item Price has a stronger linear impact (0.45) on total line-item Revenue compared to individual order Quantity (0.16). This helps us understand how pricing variations influence overall sales data.

Price Patterns: It helps us verify how individual item pricing scales impact the final calculated revenue targets.

Data Validation: This chart proves that our calculated features are mathematically correct and behave exactly as expected.

Outlier Profiling via Box Plots:



The above slide shows Box Plots for our data attributes (Quantity and Price).

In simple words, a box plot splits data into sections to easily show us normal values versus data points that are weird or completely out of place.

- Key Findings from the Chart:

Finding Data Errors: The charts show that a few specific customer orders have extreme, massive quantities compared to a normal single customer order.

Wholesale Identification: These extreme spikes point directly to bulk wholesale purchases or massive commercial returns in the store's logs.

Isolating Bad Data: By checking these graphs, we can easily set safe boundaries to separate normal customer transactions from extreme outliers before building predictive models.

Analytical Summary & Key Results:

- Core Achievements:

We successfully built a clean data pipeline to process over **525,000** transaction records.

All raw data types, missing IDs, and text date attributes were successfully fixed and cleaned.

Key Insights Uncovered:

Seasonal Cycles: Our trend lines proved that store revenue builds up steadily over the months and peaks sharply in late winter.

Feature Interactions: The correlation matrix verified that item Price acts as a stronger relative linear driver for line-item revenue compared to individual transaction Quantity.

Data Distribution: Our box plots successfully exposed extreme wholesale outliers, giving us clear limits for future data preprocessing.